PES UNIVERSITY

SOFTWARE ENGINEERING LAB (UE24CS351A)

JIRA PROJECT DELIVERABLES

Problem Statement : #40 - Student Startup Crowdfunding Platform

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1. Problem Statement & System Overview

The project is a specialized crowdfunding platform designed specifically for university students. It allows student entrepreneurs to showcase their startup ideas and raise initial seed capital from backers such as alumni, students, and outside contributors. To avoid fraud or project abandonment, funds are held in escrow and released in milestones only after review and approval by a university faculty committee.

1.1 Identified Actors

1. Student Founder: Responsible for drafting campaign details, setting reward tiers, defining funding milestones, and submitting evidence of milestone completion.

2. Campaign Backer: Browses student initiatives, selects reward tiers based on contribution amounts, makes secure financial pledges, and tracks venture updates.

3. Faculty Review Committee: Inspects submitted milestone reports, verifies project deliverables, and signs off on fund release.

4. Payment Gateway System: Tokenizes sensitive card information, holds pledged funds in escrow sub-accounts, and executes disbursement transfers.

2. Requirements Engineering

The table below lists the core functional and non-functional requirements identified for the platform along with priority, acceptance criteria, and rationale:

ID	Type	Requirement Description	Priority	Acceptance Criteria	Rationale
FR-01	Functional	Student founders can create and publish crowdfunding campaigns with pitch media, target amounts, deadlines, and reward tiers.	High	Pass: Campaign form validates fields and publishes. Fail: Incomplete details allowed to go live.	Enables student founders to showcase ideas and raise capital.
FR-02	Functional	Campaign backers can select reward tiers and make financial pledges	High	Pass: Pledges tokenized and count	Incentivizes backer

		securely.		towards campaign goal. Fail: Pledge accepted without payment authorization.	participation and records contributions accurately.
FR-03	Functional	Founders must define sequential funding milestones, and the platform must display live funding progress.	High	Pass: Milestones sum to 100% and progress bar updates in real time. Fail: Campaign created without milestone breakdown.	Enforces accountability by breaking work into measurable stages.
FR-04	Functional	Faculty review committee can inspect uploaded milestone evidence and approve or reject tranche release requests.	High	Pass: Fund tranche unlocks only with faculty sign-off. Fail: Funds disbursed without committee approval.	Protects backers from abandoned projects and ensures quality.
FR-05	Functional	System disburses funds from escrow to the student venture in tranches upon verified milestone completion.	High	Pass: Gateway transfers tranche amount after faculty sign-off. Fail: Entire funding released as a single lump sum upfront.	Reduces financial risk by aligning money release with real progress.
NFR-01	Non-Functional	Payment processing must comply with PCI-DSS standards using tokenization and TLS 1.3 encryption.	High	Pass: Zero card details stored in plain text. Fail: Card credentials stored or transmitted unencrypted.	Required for security, legal compliance, and user trust.
NFR-02	Non-Functional	System shall maintain 99.9% uptime and handle at least 500 concurrent transactions with latency under 1.5s.	Medium	Pass: Response time < 1.5s under simulated load. Fail: Latency exceeds 1.5s or server drops requests.	Ensures platform remains stable during high traffic deadlines.

3. Use-Case Flow Specification

3.1 Use-Case: UC-08 Release Funds in Tranches upon Milestone Verification

Use Case ID	UC-08
Use Case Name	Release Funds in Tranches upon Milestone Verification
Primary Actor	Faculty Review Committee Member
Supporting Actors	Student Founder, Payment Gateway System
Preconditions	1. Campaign reached minimum funding target. 2. Founder uploaded completion evidence for current milestone. 3. Faculty member is authenticated with review privileges.

Main Flow (Success)	1. Faculty member logs in and opens the campaign review queue. 2. System presents uploaded milestone deliverables, reports, and code repository links. 3. Faculty member reviews deliverables against milestone acceptance criteria. 4. Faculty member clicks 'Approve Milestone & Release Tranche' and enters remarks. 5. System logs approval timestamp and faculty ID. 6. System notifies payment gateway to disburse tranche percentage from escrow to founder account. 7. Milestone marked as Disbursed and notifications sent to founder and backers.
Alternative Flow	4a. Incomplete / Poor Deliverables: 1. Faculty member clicks 'Request Revision' and enters specific feedback. 2. Milestone status changes to 'Under Revision'. 3. No funds released until updated evidence is submitted and re-reviewed.
Postconditions	Authorized tranche released from escrow to founder; project moves to next milestone stage.

4. Jira Project Configuration & Agile Work Breakdown

The project was created in Jira Software using the Scrum template with Project Key SSCP (Student Startup Crowdfunding Platform). Five Epics were defined to represent major modules, and ten User Stories were mapped under them with estimated Story Points and priority levels.

Key	Type	Summary	Parent Epic	SP	Priority	Status
SSCP-1	Epic	Campaign Creation & Management	-	-	Medium	In Progress
SSCP-2	Epic	Backer & Reward Management	-	-	Medium	In Progress
SSCP-3	Epic	Milestone & Funding Management	-	-	Medium	In Progress
SSCP-4	Epic	Faculty Review & Fund Release	-	-	Medium	In Progress
SSCP-5	Epic	Payment & Security	-	-	Medium	In Progress
SSCP-7	Story	Create Crowdfunding Campaign	Campaign Creation & Management	5	High	Done
SSCP-8	Story	Edit/View Campaign Details	Campaign Creation & Management	3	Medium	Done
SSCP-9	Story	Select Reward Tier	Backer & Reward Management	3	High	Done
SSCP-10	Story	Manage Backer Pledge	Backer & Reward Management	5	High	Done
SSCP-11	Story	Define Funding Milestones	Milestone & Funding Management	5	High	Done
SSCP-12	Story	Track Campaign Progress	Milestone & Funding Management	3	Medium	Done
SSCP-13	Story	Review Milestone Evidence	Faculty Review & Fund Release	5	High	Done
SSCP-14	Story	Release Funds in Tranches	Faculty Review & Fund Release	8	High	Done
SSCP-15	Story	Process Backer Payment	Payment & Security	8	Highest	Done
SSCP-16	Story	Secure Payment Authorization	Payment & Security	5	High	Done

5. Sprint Reflection and Discussion

1. Did your estimates reflect the actual effort?
Ans) Yes, to a good extent. The story points helped us estimate the difficulty of the tasks before starting the sprint. Simple tasks like viewing campaign details had fewer points, while tasks like fund release and payment processing had more points because they involved more work and security considerations.

2. Was your backlog well-prioritized?
Ans) Yes. The important features were given higher priority so that the main functionality could be handled first. Campaign creation, pledges, and milestone management were considered important, followed by faculty review, fund release, and payment security features.

3. How did your simulated sprint align with your plan?
Ans) The simulated sprints followed the planned order of development. The first sprint mainly covered campaign and backer-related features, while the second sprint focused more on campaign progress, milestone review, fund release, and payment-related features. This made the work easier to organize.

4. What insights did the burndown chart give about your team's capacity?
Ans) The burndown chart helped us see how the remaining story points changed during the sprint. It showed how much work was completed and whether the team was able to finish the planned tasks. It also gave an idea of how much work the team could handle in one sprint.